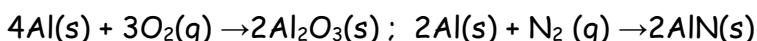


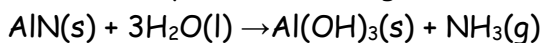
Chemical properties of aluminium

1. With air

- Strongly heated aluminium burns in air to form a white solid which is a mixture of aluminium oxide and aluminium nitride.



- The nitride yields ammonia gas when dissolved in water.

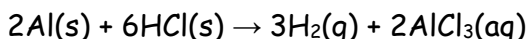


2. With water:

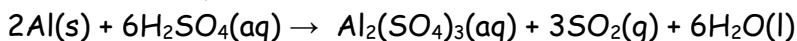
- Aluminium does not react with water. This is because of the impervious oxide layer which prevents it from further reaction.

3. With acids

- Aluminium reacts with warm moderately concentrated hydrochloric acid; aluminium chloride and hydrogen are formed.



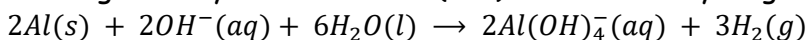
- Dilute sulphuric acid has no/little effect on aluminium due to insolubility of the protective oxide layer in the acid.
- Hot concentrated sulphuric acid oxidises aluminium to aluminium sulphate and itself reduced to sulphur dioxide and water.



- Both dilute and concentrated nitric acid render the metal passive.

4. With aqueous caustic alkalis

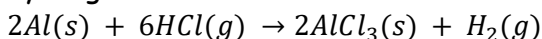
- Aluminium dissolves readily in hot concentrated solution of alkalis effervescence forming tetrahydroxoaluminate(III) in solution. Hydrogen is evolved.



- Thus, alkalis corrode aluminium. This explains why aluminium utensils (e.g. saucepans) are damaged by washing using alkaline solutions.
- Aluminium is thus, described as an amphoteric metal.

5. With non-metals

- Strongly heated, aluminium combines directly with non-metals like dry halogens, sulphur and carbon to forming AlX_3 (e.g. AlCl_3), Al_2S_3 and Al_4C_3 respectively.
- In dry current of hydrogen chloride, hot aluminium forms aluminium chloride and hydrogen.



Compounds of aluminium

- Aluminium compounds which are both ionic and covalent.
- Both aluminium oxide and aluminium fluoride are mainly ionic.

Explanation:

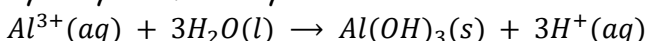
This is because both the oxide and fluoride ions have very small ionic radii. Their valence electrons are nearer to the nuclei and thus, more strongly held by their nuclei. Hence, the anions are not easily polarized by the highly charged aluminium ion.

- The other halides ($AlCl_3$, $AlBr_3$ and AlI_3) are essentially covalent.

Explanation:

This is because chloride, bromide and iodide ions have relatively larger ionic radii, their valence electrons are less strongly attracted by their nuclei. Hence, the anions are greatly polarized by the highly charged aluminium ion.

- Aqueous solutions of soluble salts of aluminium are acidic to litmus. This is due to the hydrolysis of the hydrated aluminium ions.



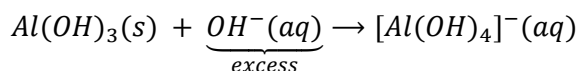
Hydrolysis occurs because the aluminium ion has high charge density/polarizing power.

(a) Aluminium oxide, Al_2O_3

- Aluminium oxide is a white solid made by strongly heating:
 - (i) aluminium powder in oxygen.
 - (ii) aluminium hydroxide, aluminium sulphate or aluminium nitrate.
- The oxide is mainly ionic, insoluble in water, melts at $2050^\circ C$ and is amphoteric.
- Aluminium oxide is used:
 - (i) as a source of aluminium and as an absorbent in column chromatography.
 - (ii) as a catalyst in organic reactions and a dehydration of ethanol to ethene.
 - (iii) as a constituent of cement and as an abrasive.

(b) Aluminium hydroxide, $Al(OH)_3$

- Aluminium hydroxide is a white gelatinous precipitate obtained when aqueous ammonia solution, sodium carbonate solution (sodium hydrogencarbonate, potassium carbonate, or potassium hydrogencarbonate) or little dilute sodium hydroxide solution is added to an aqueous solution of a soluble aluminium salt.
- The hydroxide being **amphoteric**, dissolves in strong alkalis (e.g. aqueous sodium hydroxide solution) to form a colourless solution containing aluminate ions. This is because the aluminate ions formed are soluble in water.



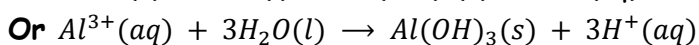
- Thus, use of strong alkalis (like sodium hydroxide solution) during its preparation should be avoided.
- The hydroxide does not dissolve in excess ammonia. This is because ammonia is a weak base, the concentration of hydroxide ions is so low and thus, soluble complex ion does not form.
- Since the hydroxide is amphoteric, it forms salts in dilute solutions with when reacted with dilute acids.

$$Al(OH)_3(s) + 3H^+(aq) \rightarrow Al^{3+}(aq) + 3H_2O(l)$$
- The hydroxide decomposes on strong heating to form aluminium oxide and steam.
- Aluminium hydroxide is used in sewage treatment since it coagulates the colloidal particles in the sewage so that they can easily be sieved. In fibres, it absorbs the dye and binds it firmly to the cloth thus, it's used as a mordant in dyeing industry.

(c) The halides

- The ionic character of the halides decrease in the order; $AlF_3 > AlCl_3 > AlBr_3 > AlI_3$
 This is due to increase in the ionic radius of the anions in the order:
 $F^- < Cl^- < Br^- < I^-$
- In fact, the chloride, bromide and iodide readily dissolve in organic solvents (e.g. benzene) and this further confirms their covalent nature.
- Except aluminium fluoride, the rest exist as dimers, Al_2X_6 in the gaseous state. The dimers dissociate to monomers at high temperatures.
- Except aluminium fluoride, the rest readily hydrolyse in water to form an acidic solution.

For example:



This explains why the halides give misty fumes/white fumes when exposed to moist air or water is added to it.

- Hydrated aluminium chloride, $AlCl_3 \cdot 6H_2O$, decomposes on heating. (Refer to the chloride of period 3 elements of the Periodic Table).
- Aluminium fluoride is sparingly soluble in water. This is due to its high lattice energy which cannot be easily offset by the hydration energy of the ions formed in aqueous solution.

$$AlF_3(s) + (aq) \rightleftharpoons Al^{3+}(aq) + 3F^-(aq)$$
- Both aluminium chloride and bromide are used as catalysts in Friedel-Crafts reactions. This is because of their ability to act as Lewis acids (i.e. electron pair acceptor).

(d) Aluminium sulphate; $Al_2(SO_4)_3 \cdot 18H_2O$

- The salt can be obtained in solution by the following reactions:
 - (i) Aluminium and hot concentrated sulphuric acid.
 - (ii) Aluminium hydroxide and warm dilute sulphuric acid.
- The resultant solution is then evaporated to dryness to obtain the crystals of aluminium sulphate.
- Aqueous solution of the salt is acidic to litmus due the hydrolysis of the hydrated aluminium ions.
 $Al^{3+}(aq) + 3H_2O(l) \rightarrow Al(OH)_3(s) + 3H^+(aq)$
- On strong heating, the sulphate decomposes leaving the oxide as a residue with formation of water vapour, sulphur dioxide and sulphur trioxide.
- The salt is used:
 - (i) in water and sewage treatment.
 - (ii) as a mordant in dyeing industry.

(e) The alums

- The alums are double salts with the general formula $M^+Y^{3+}(SO_4)_2 \cdot 12H_2O$ or $M_2SO_4 \cdot Y_2(SO_4)_3 \cdot 24H_2O$; where M^+ can be Na^+ , K^+ or NH_4^+ and Y^{3+} can be Al^{3+} , Fe^{3+} or Cr^{3+}
- Typical alums are:

Formula	Name of alum	Common name
$KAl(SO_4)_2 \cdot 12H_2O$	Aluminium potassium sulphate-12-water	Alum/potash alum
$KCr(SO_4)_2 \cdot 12H_2O$	Chromium(III) potassium sulphate-12-water	Chrome alum
$NH_4Al(SO_4)_2 \cdot 12H_2O$	Aluminium ammonium sulphate-12-water	Ammonium alum
$NH_4Fe(SO_4)_2 \cdot 12H_2O$	Ammonium iron(III) sulphate-12-water	Ferric alum

- In aqueous solution, the alums give ions. For example, potash alum solution contains K^+ , Al^{3+} and SO_4^{2-} ions
- Potash alum is used:
 - (i) in water and sewage treatment.
 - (ii) as a mordant in dyeing industry.

Tests for aluminium ions in aqueous solution.

Table showing effect of different reagents when separately treated with aqueous solution containing aluminium ions.

<i>Reagent used/added</i>	<i>Observations</i>	<i>Equations for the reactions that taking place</i>
Addition of sodium hydroxide solution	White precipitate soluble to form a colourless solution	$\text{Al}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Al}(\text{OH})_3(\text{s})$ $\text{Al}(\text{OH})_3(\text{s}) + \text{OH}^{-}(\text{aq}) \rightarrow \text{AlOH}_4^{-}(\text{aq})$
Addition of dilute ammonia solution	White precipitate insoluble in excess	$\text{Al}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Al}(\text{OH})_3(\text{s})$
Addition of sodium (potassium) carbonate solution or sodium (potassium) hydrogencarbonate solution	White precipitate and bubbles of a colourless gas	$2\text{Al}^{3+}(\text{aq}) + 3\text{CO}_3^{2-}(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Al}(\text{OH})_3(\text{s}) + 3\text{CO}_2(\text{g})$
Addition of 3-4 drops of litmus solution followed by excess dilute ammonia solution	Blue lake solution/blue solution	
Addition of 3-4 drops of alizarin solution followed by excess dilute ammonia solution	Pink lake solution/pink solution	

Trial questions

- With reference to aluminium oxide, explain what is meant by the term **amphoteric oxide**.
- Write equation for the reaction between sodium hydroxide solution and
 - silicon(IV) oxide.
 - aluminium oxide.
 - Carbon dioxide was bubbled through the solution in (a)(ii).
 - State what was observed.
 - Write equation for the reaction that took place.
- Write ionic equation for the reaction between aluminium oxide and excess sodium hydroxide solution.
 - To the resultant solution in (a) was added dilute hydrochloric acid drop wise until in excess.
 - State what was observed.
 - Write equation(s) for the reaction(s) that took place.
- Uneb paper 1; 1996 qn 12/ Uneb paper 1; 2000 qn 11/ Uneb paper 2; 2003 qn 7/ Uneb paper 2; 2012 qn 6/ Uneb paper 2; 2010 qn 7/ Uneb paper 1; 2016 qn 17

Oxidation and Reduction

A. Oxidation

The term oxidation was originally used to mean addition of oxygen to an element or a compound or removal of hydrogen from a compound.

B. Reduction

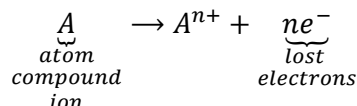
The term reduction originally meant addition of hydrogen to an element or a compound or removal of oxygen from a compound.

On the basis of the above definitions:

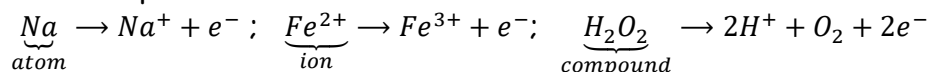
An oxidizing agent either adds oxygen or removes hydrogen while a reducing agent either adds hydrogen or removes oxygen.

The above definitions of oxidation and reduction are still valid but have been extended and are being interpreted in terms of electron transfer as follows:

Oxidation is referred to as the loss of electrons by an atom, compound or ion.



For example:

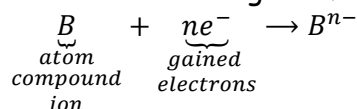


Thus, oxidation causes an increase on the charge of the original chemical species.

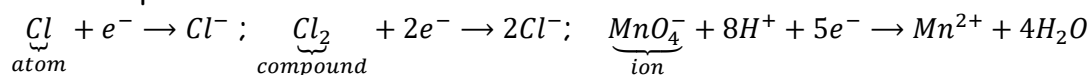
In the above reactions, Na, Fe^{2+} and H_2O_2 have been oxidised and are known as **reducing agents**.

Thus, a reducing agent is one which donates/gives electrons to another substance (i.e. it is an electron donor).

Reduction is the gain of electrons by an atom, compound or ion.



For example:



Thus, reduction causes a decrease on the charge of the original chemical species.

In the above reactions, Cl, Cl_2 and MnO_4^{-} have been reduced and are known as **oxidising agents**.

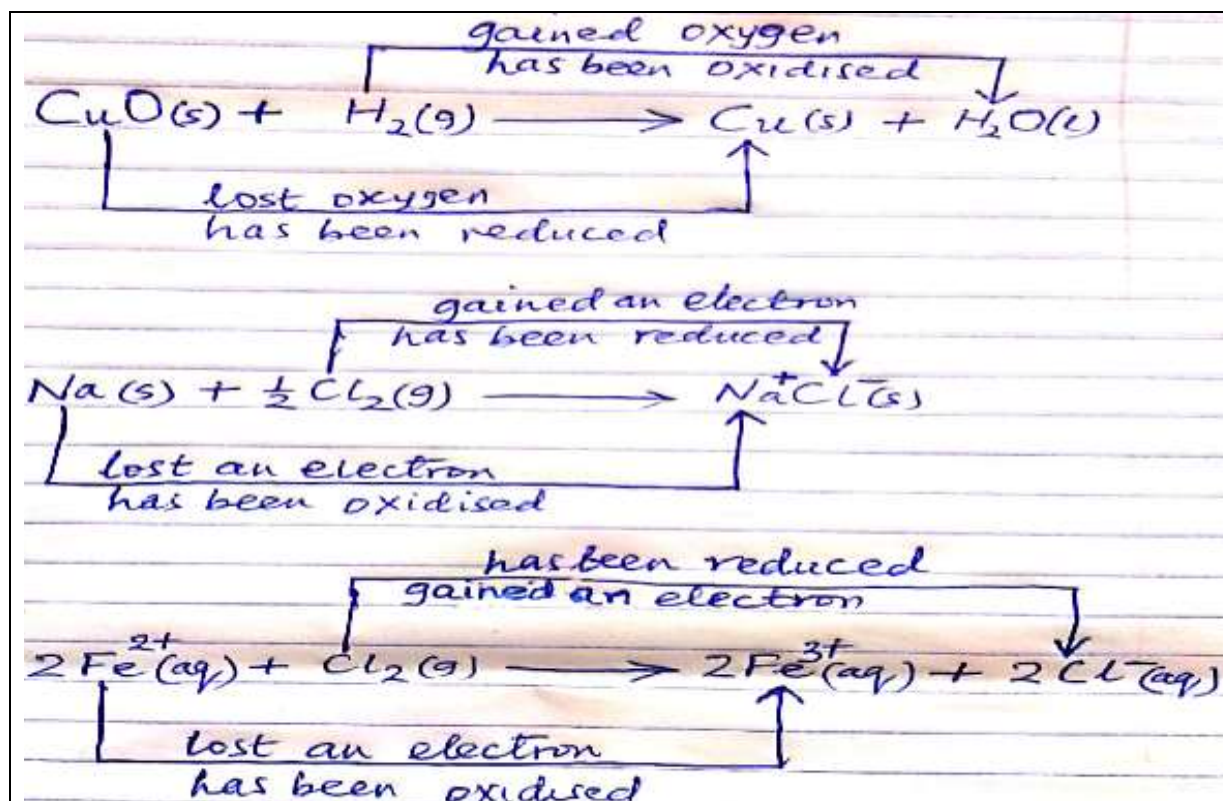
Thus, an oxidizing agent is one which accepts electrons from another substance (i.e. it is an electron acceptor).

Thus, from the above definitions, we clearly see that oxidation and reduction are complementary (i.e. occur simultaneously).

No reduction process can take place without a corresponding oxidation; that is, when a substance loses electron(s) then another must gain the electron(s).

A reaction in which there is simultaneous reduction and oxidation is termed as a **redox reaction** or **oxidation-reduction reaction**.

Examples of some redox reactions are:



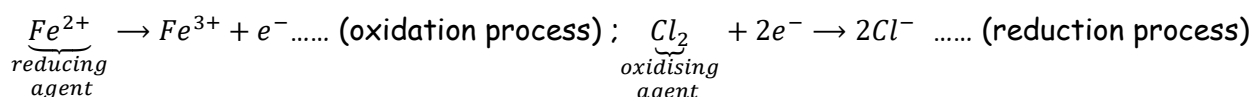
Half equations and half reactions

Oxidation and reduction are complementary processes in a redox reaction.

An oxidising agent is reduced by a reducing agent and at the same time, a reducing agent is oxidised by the oxidising agent.

Therefore, to write a redox reaction, it is useful to know how common oxidizing and reducing agents work. This can be best done by use of half equations which represent half reactions.

For example:

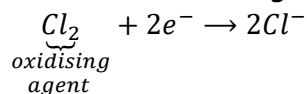


The above processes are half equations and a combination of the two half equations gives a redox reaction.

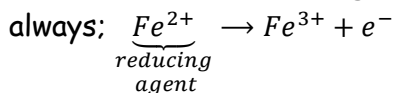
Half equation/reaction for a given reducing agent/oxidizing agent is always the same.

For example:

(a) For all oxidizing reactions involving chlorine, the half equation is always:



(b) For all reactions involving iron(II) ions as a reducing agent, the half equation is



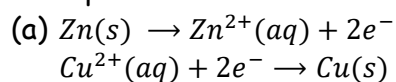
Overall equation for a redox reaction

To write a redox reaction/process, half equations are combined with elimination of electrons.

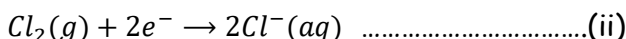
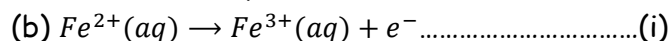
Note:

- Elimination of electrons in half equations is similar to the way we solve simultaneous equations by elimination method. The **arrow** in the half/overall equations is like an **equal sign** in simultaneous equations.
- Identical species must not appear on both sides of the arrow in the overall equations.

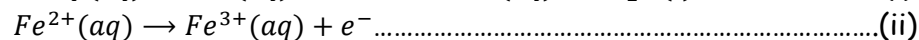
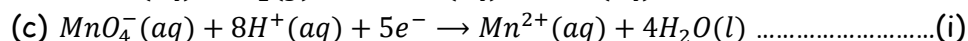
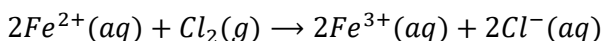
Examples:



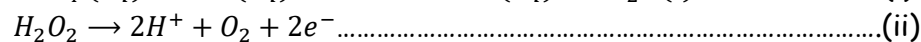
The overall equation is $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$



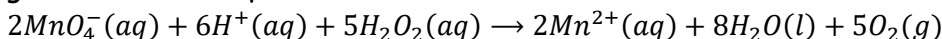
Multiplying equation (i) by 2 and adding the two half equations, the overall equation becomes:



Multiplying equation (ii) by 5 and adding the two half equations, the overall equation becomes: $\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{Fe}^{2+}(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O(l)} + 5\text{Fe}^{3+}(\text{aq})$



Multiplying equation (i) by 2 and equation (ii) by 5, then adding the two half equations gives an overall equation:



Note clearly how the number of hydrogen ions has been reduced.

Examples of oxidizing agents (oxidants)

A. Oxygen

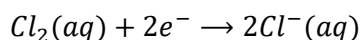
For example: $2Mg(s) + O_2(aq) \rightarrow 2MgO(s)$

B. Chlorine; Cl_2

The essential half equation is $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$

Note:

Chlorine water behaves in a similar way like gaseous chlorine. The active species in chlorine water is chlorine and thus, the essential half is:



Thus; chlorine oxidises:

(i) Iron(II) salts to iron(III) salts

Observation: Green solution turns to brown/yellow/reddish-brown solution.

Thus, we write: $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

Overall equation: $2Fe^{2+}(aq) + Cl_2(g) \rightarrow 2Fe^{3+}(aq) + 2Cl^-(aq)$

Question:

Chlorine water was added to an aqueous solution of iron(II) nitrate.

(a) State what was observed.

(b) Explain your observation in (a). Include equation for the reaction that took place.

(ii) Potassium (sodium) iodide to molecular iodine.

Observation: Colourless solution turns to reddish-brown/brown solution.

Thus, we write: $2I^-(aq) \rightarrow I_2(aq) + 2e^-$

Overall equation: $Cl_2(g) + 2I^-(aq) \rightarrow I_2(aq) + 2Cl^-(aq)$

(iii) Potassium (sodium) bromide to molecular bromine.

Observation: Colourless solution turns to a red solution.

Thus, we write: $2Br^-(aq) \rightarrow Br_2(aq) + 2e^-$

Overall equation: $Cl_2(g) + 2Br^-(aq) \rightarrow Br_2(aq) + 2Cl^-(aq)$

(iv) Hydrogen sulphide to elemental sulphur.

Observation: A yellow precipitate/solid is formed.

Thus, we write: $H_2S(g/aq) \rightarrow S(s) + 2H^+(aq) + 2e^-$

Overall equation: $H_2S(g/aq) + Cl_2(g/aq) \rightarrow S(s) + 2H^+(aq) + 2Cl^-(aq)$

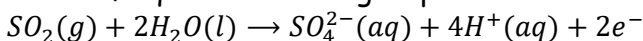
(v) Sulphur dioxide, sulphurous acid or any soluble sulphite to sulphuric acid (sulphate ion in solution).

The half equation involving sulphurous acid or any soluble sulphite is:

$SO_3^{2-}(aq) + H_2O(l) \rightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$

Overall equation: $SO_3^{2-}(aq) + H_2O(l) + Cl_2(g) \rightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2Cl^-(aq)$

The half equation involving sulphur dioxide is:



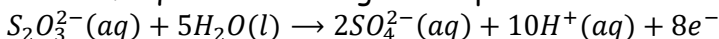
Overall equation: $SO_2(g) + 2H_2O(l) + Cl_2(g) \rightarrow SO_4^{2-}(aq) + 4H^+(aq) + 2Cl^-(aq)$

The presence of sulphuric acid (sulphate ion) can be confirmed addition of dilute nitric (hydrochloric) acid followed by barium nitrate (chloride) solution. Formation of a white precipitate confirms sulphuric acid (sulphate ion) to the solution formed.

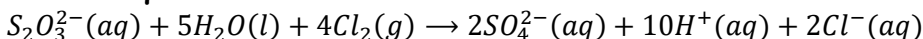
(vi) Sodium thiosulphate; $Na_2S_2O_3$

The **active** species in sodium thiosulphate is the thiosulphate ion, $S_2O_3^{2-}$.

The half equation involving thiosulphate ion is:

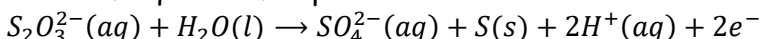


Overall equation:

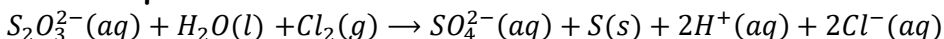


If **limited** amount of chlorine used, partial oxidation of the thiosulphate ion occurs and **insoluble** sulphur is formed. The sulphur gives rise to a **yellow precipitate**.

The half equation for partial oxidation is:



Overall equation:



Note:

Bromine or bromine water behaves in a similar way like chlorine/chlorine water. The only difference is that bromine is a weaker oxidizing agent than chlorine.

Questions:

1. When limited amount of chlorine is added into sodium thiosulphate solution, a yellow precipitate is formed. Explain this observation. Illustrate your answer with equation(s).
2. (a) Write equation for the reaction between chlorine and
(i) hydrogen sulphide. (ii) hot concentrated sodium hydroxide solution.
(b) State what was observed in (a)(i) and (a)(ii).
(c) In which of the reactions in (a) is chlorine behaving as an oxidizing agent?

C. Hydrogen peroxide; H_2O_2

In acidic medium, hydrogen peroxide acts as a strong oxidizing agent.

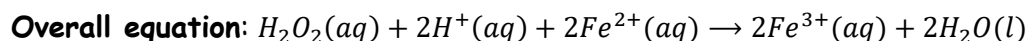
The essential half equation being: $H_2O_2(aq) + 2H^+(aq) + 2e^- \rightarrow 2H_2O(l)$

Thus; it oxidises:

(i) Iron(II) salts to iron(III) salts

Observation: Green solution turns to brown/yellow/reddish-brown solution.

Thus, we write: $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$



Questions:

1. Few drops of acidified hydrogen peroxide were added into an aqueous solution of iron(II) chloride.
 - (a) State what was observed.
 - (b) Explain your observation in (a). Include equation for the reaction that took place.
2. Write equation for the reaction between acidified iron(II) sulphate solution and hydrogen peroxide.

- (ii) Hydriodic acid or potassium (sodium) iodide to molecular iodine.

Observation: Colourless solution turns to a reddish-brown/brown solution.

Thus, we write: $2I^-(aq) \rightarrow I_2(aq) + 2e^-$

Overall equation: $H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow 2Fe^{3+}(aq) + 2H_2O(l) + I_2(aq)$

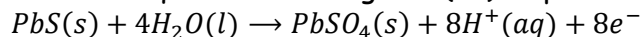
- (iii) Sulphurous acid or any soluble sulphite to sulphuric acid (sulphate ion in solution).

Overall equation is.....

- (iv) Lead(II) sulphide; PbS

The **black** lead(II) sulphide is oxidised to **white** lead(II) sulphate.

The half equation involving lead(II) sulphide is:



Overall equation is: $PbS(s) + H_2O_2(aq) \rightarrow PbSO_4(s) + 4H_2O(l)$

This reaction is used in restoring paintings in which white lead pigment/paint (lead(II) carbonate) has been blackened by hydrogen sulphide from the air.

Question:

A mixture of hydrogen peroxide and lead(II) sulphide was shaken and allowed to stand.

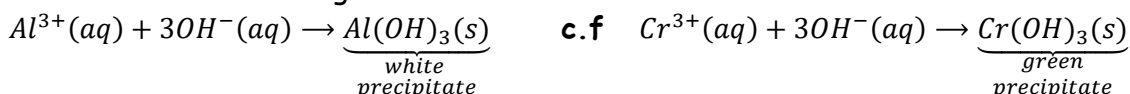
- (a) State what was observed.
- (b) Write equation for the reaction that took place.

(v) Chromium(III) hydroxide, $Cr(OH)_3$

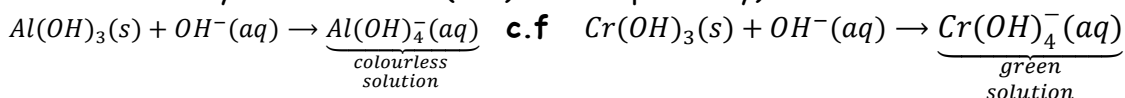
When a mixture of chromium(III) hydroxide, hydrogen peroxide and excess sodium hydroxide solution is heated/warmed, the green solution turns yellow.

Supplement:

Recall: Addition of dilute sodium hydroxide solution to a solution containing aluminium ions causes the following reaction.



Both aluminium hydroxide and chromium(III) hydroxide are amphoteric and thus, react with excess alkali to form soluble complex ions (i.e. tetrahydroxoaluminate(III) ions and tetrahydroxochromate(III) ions respectively).



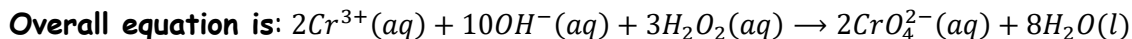
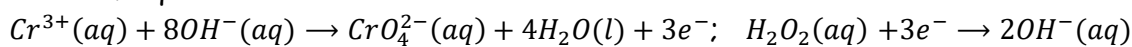
The green solution formed in excess alkali contains chromium(III) ions in form of a complex ion.

Explanation:

In presence of excess alkali, chromium(III) hydroxide being amphoteric reacts with excess hydroxide ions from the alkali to form a soluble complex ion, tetrahydroxochromate(III) ion which makes the solution green.

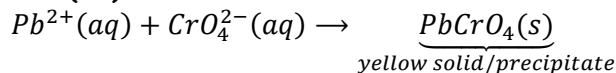
In **alkaline solution** (excess hydroxide ions), hydrogen peroxide **oxidises** chromium(III) ions in the complex ion to chromate(VI) ions which makes the solution yellow.

The half equation involved in **basic/alkaline conditions** are:



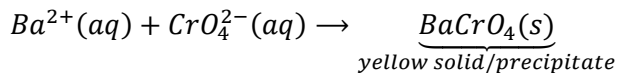
The formation of chromate(VI) ions in the yellow solution can be confirmed by addition of:

(I) lead(II) nitrate/ethanoate solution.



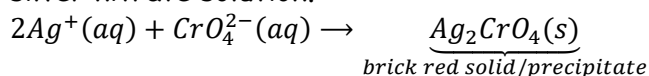
The yellow precipitate **dissolves** in excess sodium hydroxide solution to form a yellow solution. This reaction is used to confirm lead(II) ions in solution.

(II) barium nitrate/chloride solution.



The yellow precipitate is **insoluble** in excess sodium hydroxide solution. This reaction is used to confirm barium ions in solution.

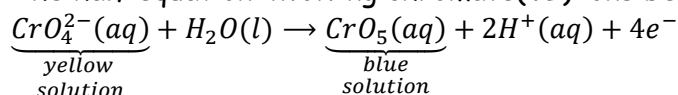
(III) silver nitrate solution.



(vi) Chromate(VI) or dichromate(VI) to a chromium peroxide, CrO_5

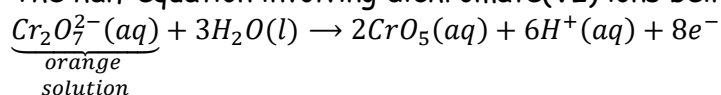
Thus the yellow/orange solution turns blue.

The half equation involving chromate(VI) ions being:



Overall equation is:

The half equation involving dichromate(VI) ions being:



Overall equation is:

Note:

The oxidizing action of hydrogen peroxide causes it to act as a bleaching agent (for hair, wax, paper pulp, cotton, silk and feathers) and as an antiseptic.

Questions:

1. A dark purple crystalline solid Z dissolves in water to form a violet. When aqueous sodium hydroxide solution was added to the solution drop wise, a green precipitate was formed, which dissolved in excess sodium hydroxide solution to give a green solution.
 - (a) Identify the cation in Z.
 - (b) Write the formula of the species responsible for the formation of the green solution.
 - (c) When hydrogen peroxide solution was added to the green solution, a yellow solution was formed which on treatment with lead(II) ethanoate solution, a yellow precipitate was formed.

Write equation for the reaction leading to the formation of

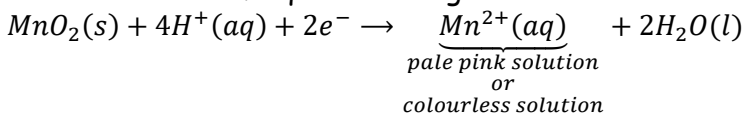
(i) green solution. (ii) yellow solution. (iii) yellow precipitate.
2. When sodium hydroxide solution was added drop wise until in excess to chromium(III) chloride solution, a green precipitate was formed that dissolved in excess sodium hydroxide solution to form a deep green solution. On addition of hydrogen peroxide solution, the solution turned from green to yellowish brown. Explain this observation. Include equation(s) for the reaction(s) that took place.
3. (a) Dilute sodium hydroxide solution was added drop wise until in excess to an aqueous chromium(III) sulphate.

- (i) State what was observed.
- (ii) Write equations for the reactions that took place.
- (b) To the product in (a) was added a few drops of hydrogen peroxide and the mixture boiled.
- (i) State what was observed.
- (ii) Write equation for the reaction that took place.

D. Manganese(IV) oxide; MnO_2

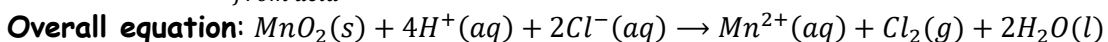
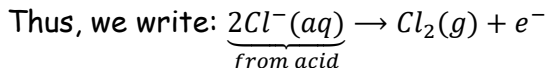
In acidic medium, the oxide is a powerful oxidizing agent.

The essential half equation being:



The oxide oxidises the following substances:

- (i) Hot concentrated hydrochloric acid to molecular chlorine.



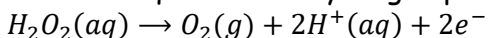
The above reaction is used in the laboratory preparation of chlorine.

Observation: Effervescence occurs and greenish-yellow gas is evolved.

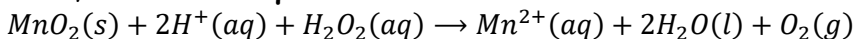
- (ii) Hydrogen peroxide to oxygen.

In this case, manganese(IV) oxide being a much stronger oxidizing agent than hydrogen peroxide, makes hydrogen peroxide act as a reducing agent.

The half equation for hydrogen peroxide acting as a reducing agent:



Thus, the **overall equation:**



This reaction is used in the laboratory preparation of oxygen.

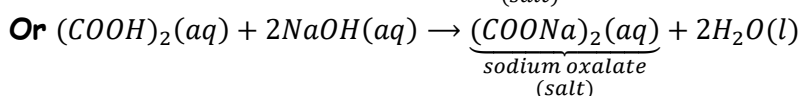
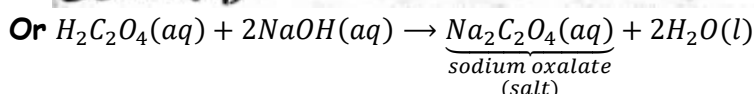
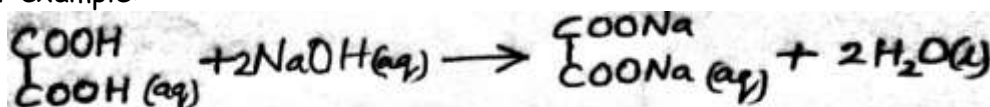
(iii) Oxalic acid (ethane-1,2-dioic acid) or oxalates (ethanedioates)

Supplement:

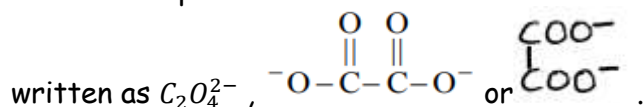
The chemical formula of oxalic acid is $H_2C_2O_4$ or $\begin{array}{c} \text{COOH} \\ | \\ \text{COOH} \end{array}$ or $\begin{array}{c} \text{O} \quad \text{O} \\ || \quad || \\ \text{HO}-\text{C}-\text{C}-\text{OH} \end{array}$.

The acid forms salts called oxalates:

For example:



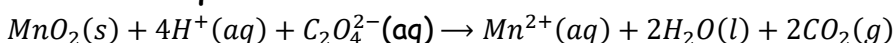
The active species in oxalic acid/the oxalates is the oxalate ion/ethanedioate ion,



The half equation for its action is $C_2O_4^{2-}(aq) \rightarrow 2CO_2(g) + 2e^-$

Thus, when oxalic acid or one of its soluble salts in solution is heated/boiled with manganese(IV) oxide in the presence of dilute sulphuric acid (to acidify the mixture/provide acidic medium), carbon dioxide is evolved.

The **overall equation** becomes:



Question:

1.55 g of an acid, $C_nH_{2n}(COOH)_2$ was dissolved in water and the resultant solution made up to 250 cm^3 .

25.0 cm^3 of the resultant solution required 23.5 cm^3 of a 0.1 M sodium hydroxide solution for complete neutralization using phenolphthalein indicator.

(a) Calculate:

(i) the molar concentration of the acid. (ii) the molecular mass of the acid.

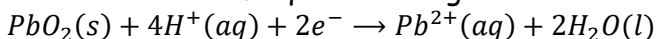
(b) Determine the value of n in the acid.

(c) Write the structural formula of the acid.

E. Lead(IV) oxide; PbO_2

In acidic medium, the oxide is a powerful oxidizing agent just like manganese(IV) dioxide.

The essential half equation being:



The oxide oxidises the following substances:

- (i) Hot concentrated hydrochloric acid to molecular chlorine.

This reaction too is used in the laboratory preparation of chlorine.

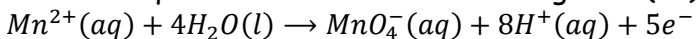
- (ii) Hydrogen peroxide to oxygen.

- (iv) Manganese(II) salts to manganate(VII) ion, MnO_4^- .

This reaction is carried out in the presence of concentrated nitric acid and the mixture is heated/boiled.

Observation: The colourless/pale pink solution turns to a purple solution.

The half equation for conversion of manganese(II) salts to manganate(VII) ion is:



Thus, the **overall equation** is:.....

This reaction is used in the laboratory as a confirmatory test for manganese(II) ions.

- (v) Sulphur dioxide to sulphur trioxide and itself reduced to lead(II) oxide when the mixture is warmed.

The lead(II) oxide formed reacts with sulphur trioxide to form **lead(II) sulphate**.

Observation: The mixture glows and a white solid is formed.

Equations: $PbO_2(s) + SO_2(g) \rightarrow PbO(s) + SO_3(g)$; $PbO(s) + SO_3(g) \rightarrow PbSO_4(s)$

Questions:

1. When lead(IV) oxide was added to a solution containing manganese(II) sulphate and concentrated nitric acid and the mixture boiled, the colour of the solution changed from pale pink to purple. Explain this observation. Illustrate your answer with equation(s).
2. Sulphur dioxide was passed over warm lead(IV) oxide.
(a) State what was observed. (b) Write equation for the reaction.

F. Iron(III) salts

The half equation for its action is: $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}$

Thus, the yellow/brown solution turns to a green solution.

The salts oxidise the following substances:

- (i) Hydrogen sulphide to elemental sulphur.

Observation: A yellow precipitate/solid is formed. Brown solution turns green.

Overall equation is:.....

- (ii) Sulphur dioxide or sulphurous acid to sulphuric acid (or sulphate ion in solution).

Observation: Brown solution turns green.

Overall equation is:.....

- (iii) Tin(II) salts to tin(IV) salts.

Observation: Brown solution turns to a green solution.

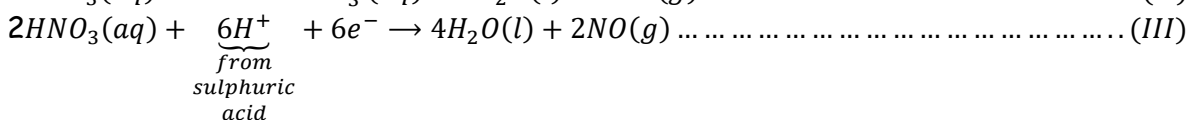
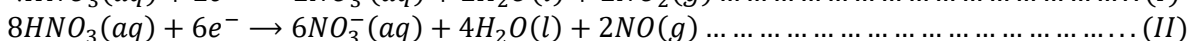
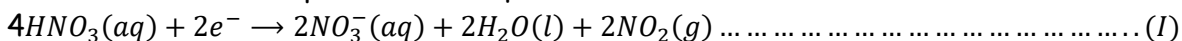
Thus, we write: $Sn^{2+}(aq) \rightarrow Sn^{4+}(aq) + 2e^-$

- Overall equation is:
- (iv) Acidified potassium iodide solution to molecular iodine.
Overall equation is:
- (v) Zinc to zinc salts.
Observation: Brown solution turns to a green solution.
Thus, we write: $\text{Zn}(s) \rightarrow \text{Zn}^{2+}(aq) + 2e^{-}$
Overall equation is:

G. Nitric acid; HNO_3

The oxidizing behaviour of this acid is quite complex due to formation of a variety of products.

Three of the more important half equations are:



The reactions above depend on the concentration of the acid, temperature at which the reaction is occurring and the nature of the reducing agent,

- Equation (I) is prominent when nitric acid is **concentrated** and reacting with copper, hydrogen sulphide or sulphur dioxide (or sulphurous acid/a soluble sulphite).
The half equation involving copper being: $\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2e^{-}$
Observation: Brown dissolves with effervescence. Green solution is formed and brown fumes evolved.
Overall equation is:
- Equation (II) is prominent when nitric acid is **dilute** and reacting with copper.
The half equation being: $\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2e^{-}$
- Equation (III) is prominent when concentrated nitric acid reacts with iron(II) salts in the presence of **dilute** sulphuric acid and heat.
The half equation being: $\text{Fe}^{2+}(aq) \rightarrow \text{Fe}^{3+}(aq) + e^{-}$
Observation: Green solution turns a brown/yellow solution. Brown fumes evolved.
Overall equation is:

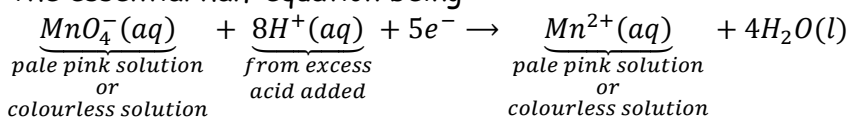
H. Potassium manganate(VII); KMnO_4

An aqueous solution of potassium manganate(VII) is purple due to manganate(VII) ions in solution.

In the presence of **sufficient** dilute sulphuric acid, the manganate(VII) ion is a powerful oxidizing agent.

Manganate(VII) ions are reduced to manganese(II) ions.

The essential half equation being:



Thus, the compound oxidises the following substances:

- (i) Concentrated hydrochloric acid to molecular chlorine in the **cold**.

Observation: Effervescence occurs and greenish-yellow gas is evolved.

Overall equation is:

The above reaction is used in the laboratory preparation of chlorine.

Note:

- The above reaction **reduces** the concentration of manganate(VII) ions in solution. This explains why aqueous solutions of potassium manganate(VII) is never acidified using hydrochloric acid.
- Similarly aqueous solutions of potassium manganate(VII) is never acidified using nitric acid. This is because both are strong oxidizing agents. Thus, there will be competition for reducing agents in the reaction mixture.

- (ii) Hydrogen peroxide to oxygen.

Again, potassium manganate(VII) being a much stronger oxidizing agent than hydrogen peroxide, makes hydrogen peroxide act as a reducing agent.

Observation: Bubbles of a colourless gas and purple solution turns colourless.

Thus, the **overall equation is**.....

- (iii) Iron(II) salts to iron(III) salts.

- (iv) Hydriodic acid or potassium (sodium) iodide to molecular iodine.

Observation: Purple solution turns to a reddish-brown/brown solution.

Overall equation is:

- (v) Tin(II) salts to tin(IV) salts.

Observation: Purple solution turns to a colourless solution.

Thus, we write: $\text{Sn}^{2+}(\text{aq}) \rightarrow \text{Sn}^{4+}(\text{aq}) + 2\text{e}^-$

Overall equation is:

- (vi) Hydrogen sulphide (or any soluble sulphide in solution) to elemental sulphur.

Observation: Purple solution turns to a colourless solution and a yellow precipitate/solid is formed.

Thus, we write: $\text{H}_2\text{S}(\text{g}) \rightarrow \text{S}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$

Overall equation is:

For soluble salts e.g. sodium/potassium/ammonium sulphide the half equation is:

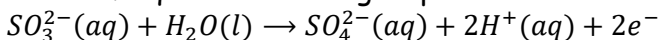
Thus, we write: $\text{S}^{2-}(\text{aq}) \rightarrow \text{S}(\text{s}) + 2\text{e}^-$

Overall equation is:

- (vii) Sulphur dioxide, sulphurous acid or any soluble sulphite to sulphuric acid (sulphate ion in solution).

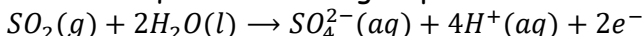
Observation: In all cases, the purple solution turns to a colourless solution.

The half equation involving sulphurous acid or any soluble sulphite is:



Overall equation is:

The half equation involving sulphur dioxide is:

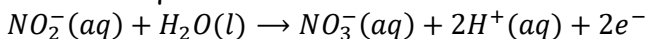


Overall equation is:

This reaction is used in the laboratory to test for sulphur dioxide gas.

- (viii) Nitrites to nitrates.

The half equation involved is:



Overall equation is:

Observation: Purple solution turns to a colourless solution.

- (ix) Oxalic acid (ethane-1,2-dioic acid) or oxalates (ethanedioates) to carbon dioxide.

Observation: Purple solution turns to a colourless solution and bubbles of a colourless gas.

This reaction is kinetically slow at temperatures below 60°C. Thus, the purple colour takes long to be discharged after each addition of the acidified potassium manganate(VII). In order to speed up the reaction rate, the oxalic acid or oxalate solution should be warmed/heated to slightly above 60°C.

Questions:

1. (a) Acidified potassium manganate(VII) reacts with ethanedioic acid.
Write the:
 - (i) half equations for the reaction. (ii) the overall equation for the reaction.
- (b) 20.00 cm³ of a 0.01 M manganate(VII) ion solution required exactly 16.65 cm³ of a solution containing 4.8 g l⁻¹ of an oxalate (COO⁻X⁺)₂.H₂O. Calculate the atomic mass of X.
2. (a) Iron(II) sulphate is normally used to standardize a solution of potassium manganate(VII) acidified using sulphuric acid.
 - (i) Write equation for the reaction between iron(II) sulphate and potassium manganate(VII).
 - (ii) Briefly explain why nitric acid and hydrochloric acid are not used to acidified potassium manganate(VII) solution.
- (b) 25 cm³ of an acidified solution of 0.02 M potassium manganate(VII) reacted exactly with 25 cm³ of sodium nitrite.
Potassium manganate(VII) reacts with sodium nitrite according to the following equation.

$$2MnO_4^-(aq) + 5NO_2^-(aq) + 6H^+(aq) \rightarrow 2Mn^{2+}(aq) + 5NO_3^-(aq) + 3H_2O(l)$$

- Calculate the concentration of the sodium nitrite in moles per litre.
- Give reasons for each of the following procedures used in analytical chemistry. Where necessary illustrate your answers with equations.
 - Potassium manganate(VII) solutions should never be acidified using hydrochloric acid.
 - A mixture of potassium manganate(VII) and sodium oxalate should be heated to temperatures above 60°C.
 - When 0.45 g of a compound Q made up of carbon, oxygen and hydrogen atoms only was burnt, 0.44 g of carbon dioxide and 0.09 g of water was formed.
 - Determine the molecular formula of Q if its relative molecular mass is 90.
 - Compound Q reacts with magnesium powder liberating hydrogen. Write the structural formula of Q.
 - Acidified manganate(VII) solution was added to aqueous solution of Q.
 - State what was observed.
 - Write equation for the reaction that took place.

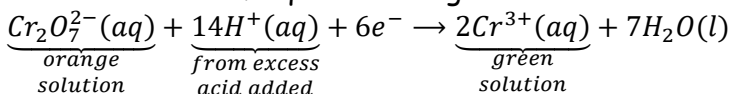
I. Potassium dichromate(VI); $K_2Cr_2O_7$

An aqueous solution of potassium dichromate(VI) is orange due to chromate(VI) ions in solution.

In aqueous solutions acidified using hydrochloric acid or sulphuric acid, the chromate(VI) ion is a very useful oxidizing agent.

Dichromate(VI) ions (which are orange) are reduced to chromium(III) ions (which are green).

The essential half equation being:



Thus, the compound oxidises the following substances:

- Iron(II) salts to iron(III) salts.

Observation: Orange solution turns to a green solution.

Note:

The change from green to yellow/brown will not be visible against the more marked colour change from orange (golden yellow) to green.

Thus, the **overall equation** is.....

- Hydriodic acid or potassium (sodium) iodide to molecular iodine.

Observation: Orange solution turns to reddish-brown/brown solution.

Note: The change from orange to green will not be visible against the more marked colour change from orange (golden yellow) to brown.

Overall equation is:

(iii) Tin(II) salts to tin(IV) salts.

Observation: Orange solution turns to a green solution.

Overall equation is:

(iv) Hydrogen sulphide (or any soluble sulphide in solution) to elemental sulphur.

Observation: Orange solution turns to a green solution and a yellow precipitate/solid is formed.

Thus, we write: $H_2S(g) \rightarrow S(s) + 2H^+(aq) + 2e^-$

Overall equation is:

For soluble salts e.g. sodium/potassium/ammonium sulphide the half equation is:

Thus, we write: $S^{2-}(aq) \rightarrow S(s) + 2e^-$

Overall equation is:

(v) Sulphur dioxide, sulphurous acid or any soluble sulphite to sulphuric acid (sulphate ion in solution).

Observation: In all cases, the orange solution turns to a green solution.

The half equation involving sulphurous acid or any soluble sulphite is:

$SO_3^{2-}(aq) + H_2O(l) \rightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$

Overall equation is:

The half equation involving sulphur dioxide is:

$SO_2(g) + 2H_2O(l) \rightarrow SO_4^{2-}(aq) + 4H^+(aq) + 2e^-$

Overall equation is:

This reaction is used in the laboratory to test for sulphur dioxide gas.

(vi) Nitrites to nitrates.

Observation: Orange solution turns to a green solution.

Overall equation is:

J. Halic(V) acid and their salts halates(V); e.g. $HClO_3$, HIO_3 , $KClO_3$, KIO_3 and $NaClO_3$

They contain halate(V) ions, XO_3^- which in acidic medium are strongly oxidizing.

For example; iodate(V) ion and chlorate(V) ions.

The essential half equation being:

$\underline{XO_3^-(aq)} + 6H^+(aq) + 6e^- \rightarrow X^-(aq) + 3H_2O(l)$; X= Cl or I

colourless solution

Thus, they oxidise iron(II) salts, sulphur dioxide (or sulphites), hydriodic acid (or iodide ions) and hydrogen sulphide.

Examples of reducing agents (reductants)

A. Metals

$X(s) \rightarrow X^+(aq) + ne^-$; n is the valency of X

B. Hydrogen peroxide

Hydrogen peroxide reduces lead(IV) dioxide, manganes(IV) oxide, iron(III) salts, silver(I) oxide and manganate(VII) ions.

C. Hydrogen sulphide

Hydrogen sulphide reduces sulphur dioxide, halogens, hydrogen peroxide, concentrated nitric acid, iron(III) salts, dichromate(VI) ions and manganate(VII) ions.

D. Iron(II) salts

The half equation is: $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

Iron(II) salts reduce bromine/chlorine, hydrogen peroxide, concentrated nitric acid, dichromate(VI) ions and manganate(VII) ions.

E. Sulphur dioxide/sulphurous acid (or any soluble sulphite)

They reduce iron(III) salts, bromine/chlorine, hydrogen peroxide, concentrated nitric acid, dichromate(VI) ions and manganate(VII) ions.

F. Oxalic acid/any soluble oxalate

They reduce manganese(IV) oxide, dichromate(VI) ions and manganate(VII) ions.

G. Hydriodic acid/acidified potassium iodide solution

The half equation is: $2I^-(aq) \rightarrow I_2(aq) + 2e^-$

They reduce iron(III) salts, hydrogen peroxide, concentrated nitric acid, dichromate(VI) ions and manganate(VII) ions.

H. Tin(II) salts

The half equation is: $Sn^{2+}(aq) \rightarrow Sn^{4+}(aq) + 2e^-$

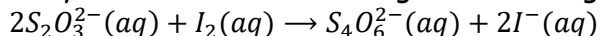
They reduce iodine, iron(III), dichromate(VI) ions and manganate(VII) ions.

I. Sodium thiosulphate

The active species in sodium thiosulphate is the thiosulphate ion, $S_2O_3^{2-}$; which is oxidised to tetrathionate ion, $S_4O_6^{2-}$

The half equation for its action is: $2S_2O_3^{2-}(aq) \rightarrow S_4O_6^{2-}(aq) + 2e^-$

They reduce iodine according the following equation.



Thus, the colourless solution turns to a brown solution.

Trial Questions:

1. Write equation for the reaction between:

- acidified potassium dichromate(vi) and sodium nitrite.
- potassium chromate and barium chloride solution.
- (i) dilute nitric acid and copper. (ii) concentrated nitric acid and copper.
- sulphite ions and iron(III) chloride.
- sulphite ions and potassium dichromate in acid solution.

2. (a) State what would be observed and write equation for the reaction that would

take place if:

- (i) excess hydrogen peroxide was added to acidified solution of potassium manganate(VII).
- (ii) hydrogen peroxide was added to acidified solution of potassium dichromate(VI) solution.
- (iii) a solution of hydrogen sulphide was added to acidified potassium dichromate(VI) solution.
- (iv) hydrogen sulphide gas was bubbled through an acidified solution of potassium manganate(VII).
- (v) tin(II) chloride was added to acidified solution of potassium dichromate(VI) solution.
- (vi) a mixture of acidified solution of potassium manganate(VII) and ethane-1,2-dioic acid is heated.
- (vii) tin(II) chloride was added to iron(III) sulphate solution.
- (viii) copper was added to a solution of concentrated nitric acid.
- (ix) potassium iodide is added to acidified solution of hydrogen peroxide.
- (x) sodium sulphite is added to a solution of acidified of potassium dichromate(VI).
- (xi) aqueous iron(II) sulphate is added to acidified potassium manganate(VII) solution.
- (xii) dilute sulphuric acid is added to a solution containing iodate and iodide ions.
- (xiii) sulphur dioxide was bubbled through acidified potassium dichromate(VI) solution.

(b) Give a reason for your observation in (a)(i) above.

3. When hydrazine (NH_2NH_2) is added to an acidified potassium manganate(VII) solution, the potassium manganate(VII) solution turns from purple to colourless and a colourless gas is evolved. Explain this observation. Illustrate your answer with equations
4. 3.8 g of solder containing tin was dissolved in excess hydrochloric acid. The solution was made up to 250 cm^3 . 25.0 cm^3 of this solution required 23.5 cm^3 of a 0.01 M potassium dichromate(VI) solution for complete reaction.
- (a) Write the half equation for potassium dichromate(VI) acting as an oxidizing agent in acid medium.
- (b) Calculate the number of moles of:
- (i) potassium dichromate(VI) used.
 - (ii) tin in the 250 cm^3 of solution.
- (c) Determine the percentage, by mass of tin in the solder.
5. State the conditions and write equations for the reaction between hydrogen peroxide and:
- (a) Iron(II) ions. (b) Iron(III) ions. (c) Iodide ions.